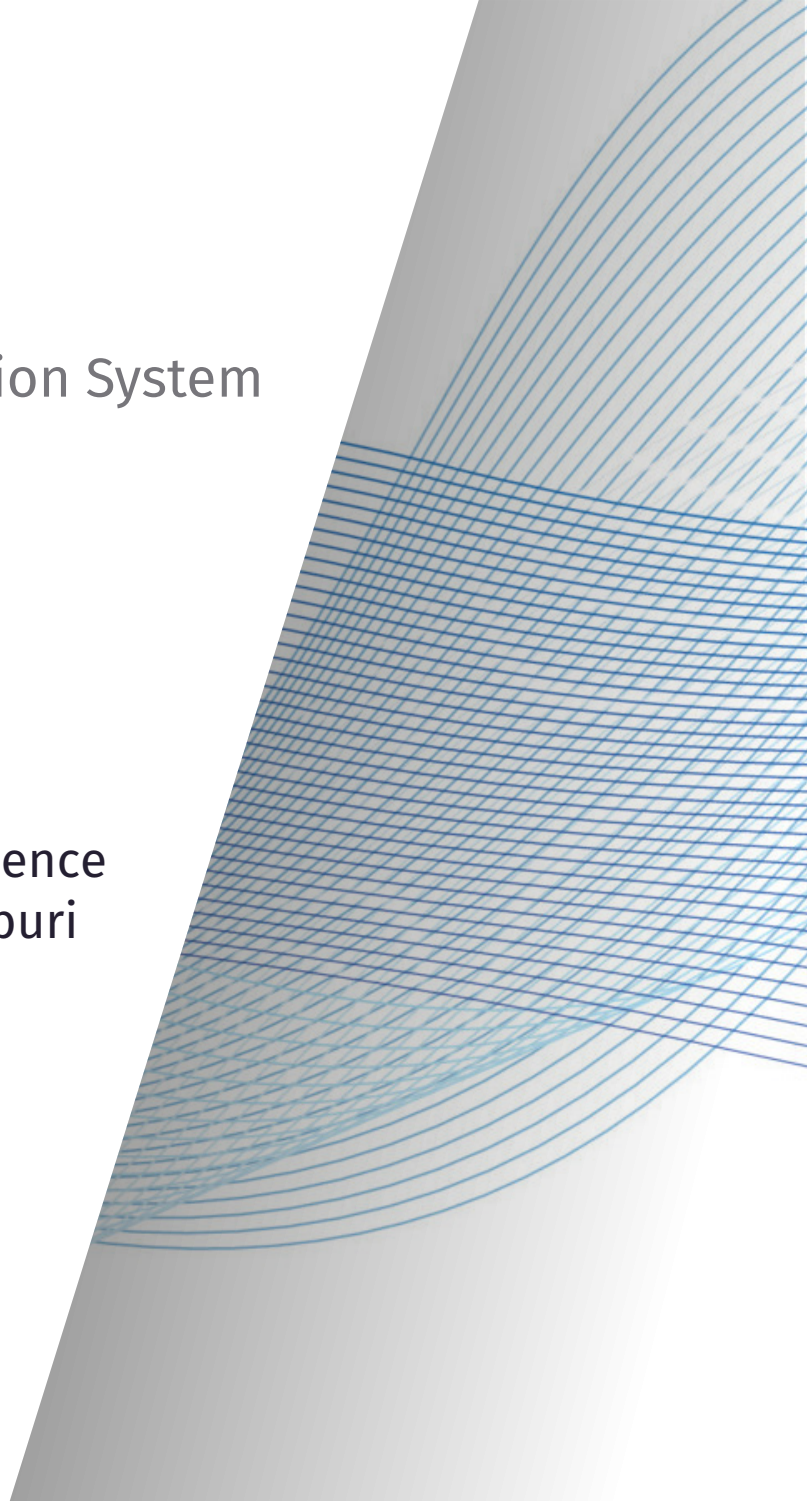
# Figures and Tables

LaTeX: A Professional Document Preparation System

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# Agenda

1. Basic Graphics
   - 1.1 Coloring
   - 1.2 Figures
   - 1.3 Cross-References
2. Tables
   - 2.1 Tabular Environment
   - 2.2 Column Aligning
   - 2.3 Cross-References

# Using Color in LaTeX

This topic explains how to use color in your LaTeX document via the `xcolor` package which provides a set of commands for color manipulation and supports a larger number of color models.

```
\usepackage{xcolor}
```

```
This is some sample {\color{red} text with color}.
We can use many {\color{blue} built-in}
 ↪  {\color{violet} colors} {\color{brown} here}.
```

## Output

This is some sample text with color. We can use many built-in colors here.

# Named colors in xcolor package






# More named colors options

`xcolor` package options:

- dvipsnames: loads 68 named colors (CMYK)
- svgnames: loads 151 named colors (RGB)
- x11names: loads 317 named colors (RGB)

```
\usepackage[dvipsnames]{xcolor}
```

| | | | |
|---|---|---|---|
| Apricot | Aquamarine | Bittersweet | Black |
| Blue | BlueGreen | BlueViolet | BrickRed |
| Brown | BurntOrange | CadetBlue | CarnationPink |
| Cerulean | CornflowerBlue | Cyan | Dandelion |
| DarkOrchid | Emerald | ForestGreen | Fuchsia |
| Goldenrod | Gray | Green | GreenYellow |
| JungleGreen | Lavender | LimeGreen | Magenta |
| Mahogany | Maroon | Melon | MidnightBlue |
| Mulberry | NavyBlue | OliveGreen | Orange |
| OrangeRed | Orchid | Peach | Periwinkle |
| PineGreen | Plum | ProcessBlue | Purple |
| RawSienna | Red | RedOrange | RedViolet |
| Rhodamine | RoyalBlue | RoyalPurple | RubineRed |
| Salmon | SeaGreen | Sepia | SkyBlue |
| SpringGreen | Tan | TealBlue | Thistle |
| Turquoise | Violet | VioletRed | White |
| WildStrawberry | Yellow | YellowGreen | YellowOrange |

# Defining your own colors

```
\usepackage{xcolor}

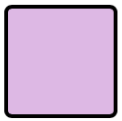
\definecolor{mypink1}{rgb}{0.858, 0.188, 0.478}
\definecolor{mypink2}{RGB}{219, 48, 122}
\definecolor{mypink3}{cmyk}{0, 0.7808, 0.4429, 0.1412}
\definecolor{mypink4}{HTML}{F88379}
```

```
Defined color with {\color{mypink1} different color models}.
Defined color with {\color{mypink2} different color models}.
Defined color with {\color{mypink3} different color models}.
Defined color with {\color{mypink4} different color models}.
```

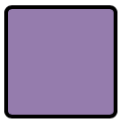
## Output

Defined color with different color models.
Defined color with different color models.
Defined color with different color models.
Defined color with different color models.

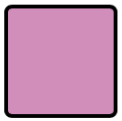
# Color Codes



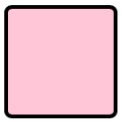
Name: Thistle
Hex: #E0BBE4
RGB: (224, 187, 228)
CMYK: 0.017, 0.179, 0, 0.105



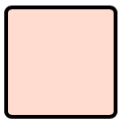
Name: Lavender Purple
Hex: #957DAD
RGB: (149, 125, 173)
CMYK: 0.138, 0.277, 0, 0.321



Name: Pastel Violet
Hex: #D291BC
RGB: (210, 145, 188)
CMYK: 0, 0.309, 0.104, 0.176



Name: Cotton Candy
Hex: #FEC8D8
RGB: (254, 200, 216)
CMYK: 0, 0.212, 0.149, 0.003



Name: Lumber
Hex: #FFDFD3
RGB: (255, 223, 211)
CMYK: 0, 0.125, 0.172, 0

## Find more color codes and schemes:

- https://www.schemecolor.com
- https://colorhunt.co
- https://htmlcolorcodes.com
- https://paletton.com
- https://coolors.co
- https://mycolor.space
- http://colormind.io
- https://color.adobe.com/create/color-wheel
- https://www.canva.com/colors/color-palettes/

# Mixing colors

```
\usepackage{xcolor}

\colorlet{LightRed}{red!50}
\colorlet{CottonCandy}{pink!80!blue}
```

```
This is {\color{LightRed} how we mixed colors}.

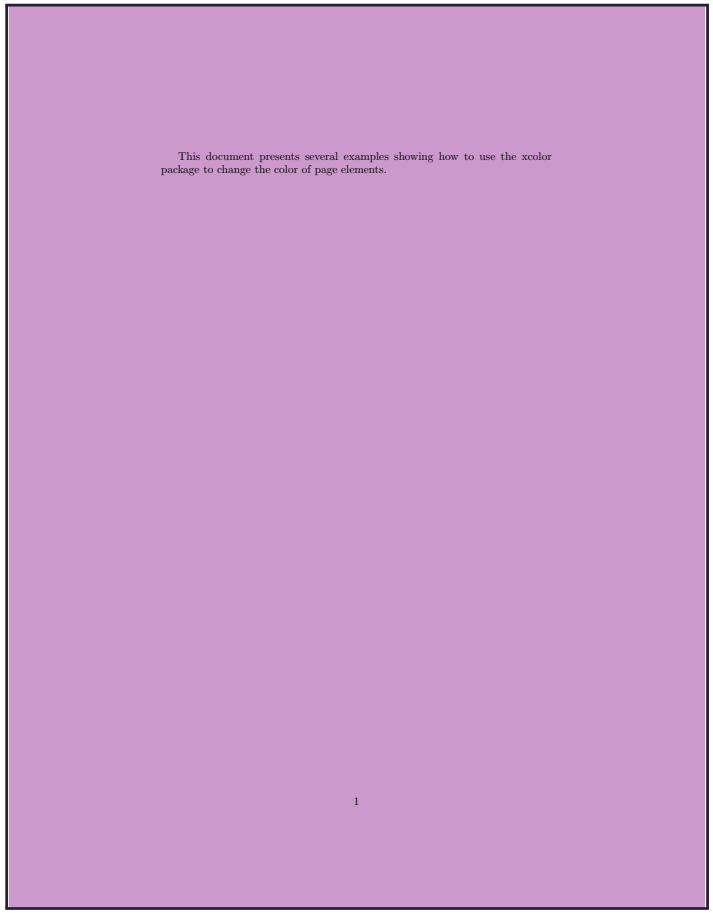
This is {\color{CottonCandy} how we mixed colors}.
```

## Output

This is how we mixed colors.
This is how we mixed colors.

# Setting the page background color

```
1  \documentclass{article}
2  \usepackage{xcolor}
3  \colorlet{CottonCandy}{pink!80!blue}
4
5  \begin{document}
6  \pagecolor{CottonCandy}
7  This document presents several
   ↪  examples showing how to use
   ↪  the xcolor package to change
   ↪  the color of page elements.
8  \end{document}
```



# Inserting Images

```
1  \documentclass{article}
2  \usepackage{graphicx}
3  \graphicspath{ {./images/} }
4
5  \begin{document}
6  The universe is immense and
7  it seems to be homogeneous,
8  in a large scale,
9  everywhere we look at.
10
11 \includegraphics{universe.jpeg}
12
13 There's a picture of a galaxy
   ↪  above
14 \end{document}
```

The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.



There's a picture of a galaxy above

# Changing the image size

```
1  \documentclass{article}
2  \usepackage{graphicx}
3  \graphicspath{ {./images/} }
4
5  \begin{document}
6  The universe is immense and
7  it seems to be homogeneous,
8  in a large scale,
9  everywhere we look at.
10
11 \includegraphics[scale=0.5]
   ↪  {universe.jpeg}
12
13 There's a picture of a galaxy
   ↪  above
14 \end{document}
```

The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.



There's a picture of a galaxy above

# Changing the image size

```
1  \documentclass{article}
2  \usepackage{graphicx}
3  \graphicspath{ {./images/} }
4
5  \begin{document}
6  The universe is immense and
7  it seems to be homogeneous,
8  in a large scale,
9  everywhere we look at.
10
11 \includegraphics[width=4cm,
   ↪  height=2cm]
   ↪  {universe.jpeg}
12
13 There's a picture of a galaxy
   ↪  above
14 \end{document}
```

The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.



There's a picture of a galaxy above

# Rotating the image

```
1  \documentclass{article}
2  \usepackage{graphicx}
3  \graphicspath{ {./images/} }
4
5  \begin{document}
6  The universe is immense and
7  it seems to be homogeneous,
8  in a large scale,
9  everywhere we look at.
10
11 \includegraphics[scale=0.5,
   ↪  angle=45]{universe.jpeg}
12
13 There's a picture of a galaxy
   ↪  above
14 \end{document}
```

The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.



There's a picture of a galaxy above

# Figures

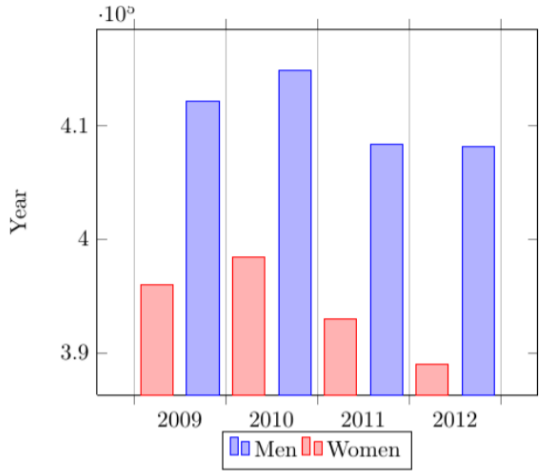
The `figure` environment is used to display pictures as **floating elements** within the document.

- This means you include the picture inside the figure environment and you don't have to worry about it's placement.
- LaTeX will position it in a such way that it fits the flow of the document.

# Figures

```
In this picture you can see a bar graph that shows
the results of a survey which involved some important
data studied as time passed.

\begin{figure}[t]
\includegraphics[width=8cm]{Plot.png}
\centering
\end{figure}
```




In this picture you can see a bar graph that shows the results of a survey which involved some tricky data studied as time passed.

# Figure's Positionings

| Parameter | Position |
|---|---|
| h | Place the float here, i.e., approximately at the same point it occurs in the source text (however, not exactly at the spot). |
| t | Position at the top of the page. |
| b | Position at the bottom of the page. |
| p | Put on a special page for floats only. |
| ! | Override internal parameters LaTeX uses for determining "good" float positions. |
| H | Places the float at precisely the location in the LaTeX code. Requires the float package, though may cause problems occasionally. This is somewhat equivalent to h!. |

# Wrapping text around figures

```
\usepackage{wrapfig}
```

```
\begin{wrapfigure}{r}{4cm} %this will be at the right
    \centering
    \includegraphics[width=4cm]{mesh.png}
\end{wrapfigure}

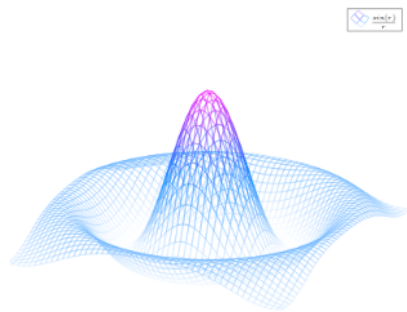
There are several ways to plot a function of two variables, depending on the
↪  information you are interested in.
For instance, if you want to see the mesh of a function so it easier to see the
↪  derivative you can use a plot like the one on the left.

\begin{wrapfigure}{l}{4cm}
    \centering
    \includegraphics[width=4cm]{contour.png}
\end{wrapfigure}

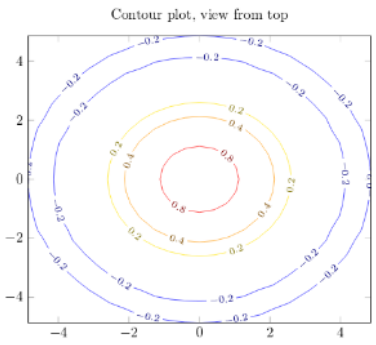
On the other side, if you are only interested on certain values you can use the
↪  contour plot, you can use the contour plot, you can use the contour plot,
↪  you can use the contour plot, you can use the contour plot, you can use the
↪  contour plot, you can use the contour plot, like the one on the left.
```

# Output

There are several ways to plot a function of two variables, depending on the information you are interested in. For instance, if you want to see the mesh of a function so it easier to see the derivative you can use a plot like the one on the left.



On the other side, if you are only interested on certain values you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, like the one on the left.



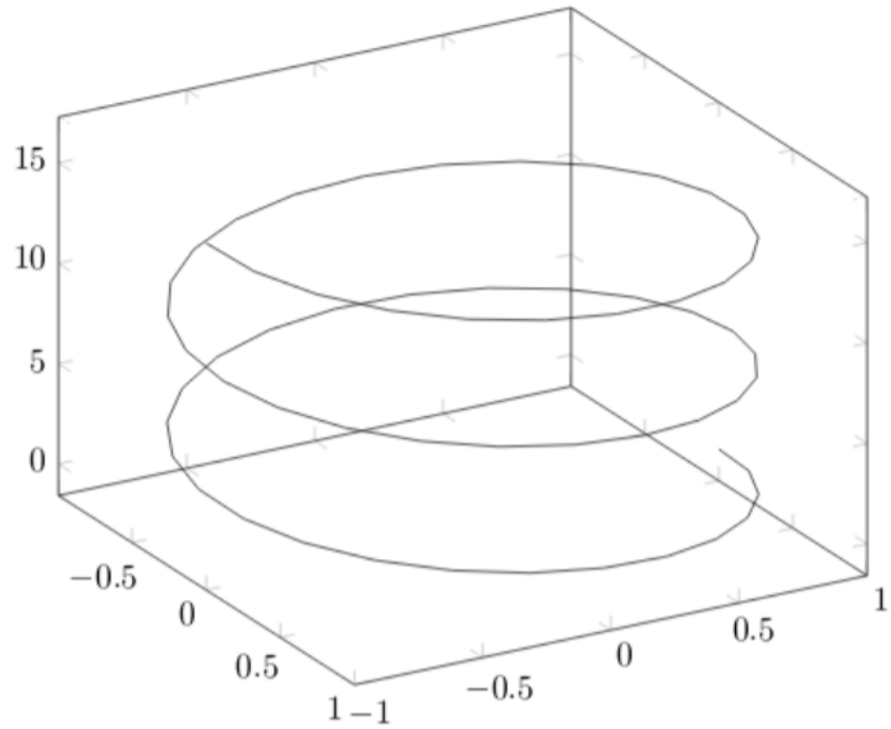

On the other side, if you are only interested on certain values you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, like the one on the left.

On the other side, if you are only interested on certain values you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, you can use the contour plot, like the one on the left.

# Figure's Captioning

```
1 \begin{figure}[h]
2 \caption{Example of a parametric plot (sin(x), cos(x), x)}
3 \centering
4 \includegraphics[width=0.5\textwidth]{spiral.png}
5 \end{figure}
```

Figure 1: Example of a parametric plot $(\sin(x), \cos(x), x)$



# Labels and Cross-references

```latex
\begin{figure}[h]
    \centering
    \includegraphics[width=0.25\textwidth]{mesh.png}
    \caption{a nice plot}
    \label{fig:mesh1}
\end{figure}

As you can see in the figure \ref{fig:mesh1}, the
function grows near 0. Also, in the page \pageref{fig:mesh1}
is the same example.
```

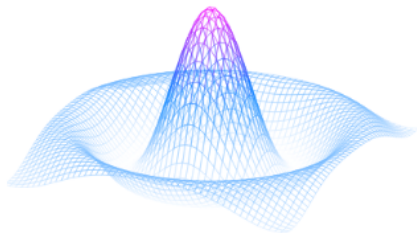


Figure 3: a nice plot

As you can see in the figure 3, the function grows near 0. Also, in the page 7 is the same example.

# List of Figures

Another great characteristic in a LaTeX document is the ability to automatically generate a list of figures. This is straightforward.

```
\listoffigures
```

## List of Figures


1 Example of a parametric plot $(\sin(x), \cos(x), x)$ . . . . . 6
2 Using again the picture of the universe. This caption will be on the right . . . . . 6
3 a nice plot . . . . . 7
4 a nice contour plot . . . . . 7


This command **only works on captioned figures**.

# Reference Guide: Length Abbrevation

| Abbrevation | Definition |
|---|---|
| pt | A point, is the default length unit. About 0.3515mm |
| mm | a millimetre |
| cm | a centimetre |
| in | an inch |
| ex | the height of an x in the current font |
| em | the width of an m in the current font |
| `\columnsep` | distance between columns |
| `\columnwidth` | width of the column |
| `\linewidth` | width of the line in the current environment |
| `\paperwidth` | width of the page |
| `\paperheight` | height of the page |
| `\textwidth` | width of the text |
| `\textheight` | height of the text |

# Reference Guide: Supported Graphic Formats

| Format | Recommendation |
|---|---|
| jpg/jpeg | Popular bitmap graphic format. Good choice if we want to insert photos. |
| png | Better quality than jpg/jpeg format and supported transparent backgrounds. However, the file size is averagely larger. |
| pdf | Best choice and quaity if it is sourced with vector graphic. |
| eps | These can be included using the `epstopdf` package (we just need to install the package, we don't need to use `\usepackage{}` to include it in our document.) |

# Tabular Environment

```
\begin{tabular}{ c c c }
 cell1 & cell2 & cell3 \\
 cell4 & cell5 & cell6 \\
 cell7 & cell8 & cell9
\end{tabular}
```

| | | |
|---|---|---|
| cell1 | cell2 | cell3 |
| cell4 | cell5 | cell6 |
| cell7 | cell8 | cell9 |

# Adding borders

```
\begin{tabular}{ |c|c|c| }
 \hline
 cell1 & cell2 & cell3 \\
 cell4 & cell5 & cell6 \\
 cell7 & cell8 & cell9 \\
 \hline
\end{tabular}
```

| | | |
|---|---|---|
| cell1 | cell2 | cell3 |
| cell4 | cell5 | cell6 |
| cell7 | cell8 | cell9 |

# Adding more borders

```
\begin{tabular}{||c c c c||}
 \hline
 Col1 & Col2 & Col2 & Col3
 ↪  \\[0.5ex]
 \hline\hline
 1 & 6 & 87837 & 787 \\
 \hline
 2 & 7 & 78 & 5415 \\
 \hline
 3 & 545 & 778 & 7507 \\
 \hline
 4 & 545 & 18744 & 7560 \\
 \hline
 5 & 88 & 788 & 6344 \\[1ex]
 \hline
\end{tabular}
```

| Col1 | Col2 | Col2 | Col3 |
|---|---|---|---|
| 1 | 6 | 87837 | 787 |
| 2 | 7 | 78 | 5415 |
| 3 | 545 | 778 | 7507 |
| 4 | 545 | 18744 | 7560 |
| 5 | 88 | 788 | 6344 |

# Column Aligning

| Abbrevation | Alignment |
|---|---|
| c | The contents of the column will be centered. |
| l | The contents of the column will be left aligned. |
| r | The contents of the column will be right aligned. |
| p{xxx} | The contents of the column will be aligned as a paragraph with fixed length of text width, e.g., `p{5cm}` or `p{0.45\textwidth}`. |

# Positioning Tables

```
1  Below is a table positioned
   ↪ exactly here:
2  \begin{table}[h!]
3  \centering
4   \begin{tabular}{||c c c c||}
5   \hline
6   Col1 & Col2 & Col2 & Col3 \\
    ↪ [0.5ex]
7   \hline\hline
8   1 & 6 & 87837 & 787 \\
9   2 & 7 & 78 & 5415 \\
10  3 & 545 & 778 & 7507 \\
11  4 & 545 & 18744 & 7560 \\
12  5 & 88 & 788 & 6344 \\ [1ex]
13  \hline
14  \end{tabular}
15 \end{table}
```

Below is a table positioned exactly here:

| Col1 | Col2 | Col2 | Col3 |
|---|---|---|---|
| 1 | 6 | 87837 | 787 |
| 2 | 7 | 78 | 5415 |
| 3 | 545 | 778 | 7507 |
| 4 | 545 | 18744 | 7560 |
| 5 | 88 | 788 | 6344 |

# Captions and Cross-references

```latex
Table \ref{table:1} is
an example of a referenced
 ↪  \LaTeX{} element.

\begin{table}[h!]
\centering
\begin{tabular}{|c|c|c|c|}
 \hline
 Col1 & Col2 & Col2 & Col3
  ↪  \\
 \hline
 1 & 6 & 87837 & 787 \\
 2 & 7 & 78 & 5415 \\
 3 & 545 & 778 & 7507 \\
 4 & 545 & 18744 & 7560 \\
 \hline
\end{tabular}
\caption{Table to test
 ↪  captions and labels.}
\label{table:1}
\end{table}
```

Table 1 is an example of a referenced LaTeX element.

| Col1 | Col2 | Col2 | Col3 |
|---|---|---|---|
| 1 | 6 | 87837 | 787 |
| 2 | 7 | 78 | 5415 |
| 3 | 545 | 778 | 7507 |
| 4 | 545 | 18744 | 7560 |

**Table 1:** Table to test captions and labels.

# List of Tables

To create a list of tables use the command. The caption of each table will be used to generate this list

```
\listoftables
```

List of Tables


1 This is the caption for the first table. . . . . 1
2 This is the caption for the second table. . . . . 1


This command **only works on captioned tables**.

# Merging Cells and Partial Lines

- `\multicolumn{n}{spec}{text}` merges $n$ cells of a row into one cell with its own column spec.
- `\cline{i-j}` draws a line under columns $i$ to $j$ only.

```
\begin{tabular}{|l|c|c|}
 \hline
 & \multicolumn{2}{c|}{Price} \\
 \cline{2-3}
 Drink & Hot & Iced \\
 \hline
 Latte & 55 & 60 \\
 Mocha & 60 & 65 \\
 \hline
\end{tabular}
```

| | Price | |
|---|---|---|
| Drink | Hot | Iced |
| Latte | 55 | 60 |
| Mocha | 60 | 65 |

The spec `c|` keeps the right border of the merged cell; the left border comes from the cell before it.

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RMUTT, November 2026

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